

Vegetarian diets and incidence of diabetes in the Adventist Health Study-2



Aim To evaluate the relationship of diet to incident diabetes among non-Black and Black participants in the Adventist Health Study-2. **Methods and Results** Participants were 15,200 men and 26,187 women (17.3% Blacks) across the U.S. and Canada who were free of diabetes and who provided demographic, anthropometric, lifestyle and dietary data. Participants were grouped as vegan, lacto ovo vegetarian, pesco vegetarian, semi-vegetarian or non-vegetarian (reference group). A follow-up questionnaire after two years elicited information on the development of diabetes. Cases of diabetes developed in 0.54% of vegans, 1.08% of lacto ovo vegetarians, 1.29% of pesco vegetarians, 0.92% of semi-vegetarians and 2.12% of non-vegetarians. Blacks had an increased risk compared to non-Blacks (odds ratio [OR] 1.364; 95% confidence interval [CI], 1.093–1.702). In multiple logistic regression analysis controlling for age, gender, education, income, television watching, physical activity, sleep, alcohol use, smoking and BMI, vegans (OR 0.381; 95% CI 0.236–0.617), lacto ovo vegetarians (OR 0.618; 95% CI 0.503–0.760) and semi-vegetarians (OR 0.486, 95% CI 0.312–0.755) had a lower risk of diabetes than non-vegetarians. In non-Blacks vegan, lacto ovo and semi-vegetarian diets were protective against diabetes (OR 0.429, 95% CI 0.249–0.740; OR 0.684, 95% CI 0.542–0.862; OR 0.501, 95% CI 0.303–0.827); among Blacks vegan and lacto ovo vegetarian diets were protective (OR 0.304, 95% CI 0.110–0.842; OR 0.472, 95% CI 0.270–0.825). These associations were strengthened when BMI was removed from the analyses. **Conclusion** Vegetarian diets (vegan, lacto ovo, semi-) were associated with a substantial and independent reduction in diabetes incidence. In Blacks the dimension of the protection associated with vegetarian diets was as great as the excess risk associated with Black ethnicity.